DANIELLE A. DICKINSON

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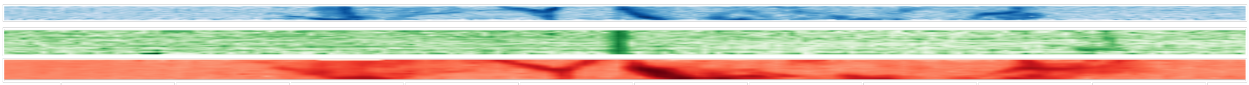
https://danielle-dickinson.github.io



Cross-sections of JWST MIRI MRS IFUs of Cassiopeia A's unshocked ejecta in [S IV] (blue), [Ne II] (green), and [O IV] (red).

I observe **mass loss in massive stars** via extragalactic studies of supernovae as well as study the **supernova remnants**, Cassiopeia A and G292. Outside of Purdue and into the outdoors, I enjoy rock climbing and running.

I am driven to create a diverse and welcoming atmosphere for all students. I am a proud member of Purdue's Gender Minority in Physics Club, and I am the president of the Physics Graduate Student Association.

Skills: Python, HTML, C, C++, Astropy, Matplotlib, Leadership, Course Development, Graphic Design, Digital Art, Performance Arts

Languages: English, Spanish

EDUCATION

Purdue University *Since August 2021*
Ph.D., Physics *(In Progress)*
M.Sc., Physics *(In Progress)*
Advised by Prof. Danny Milisavljevic
Thesis: Reconstructing the Mass Loss Environments and Explosive Debris of Core-Collapse Supernovae

University of Arizona *August 2017 - May 2021*
B.Sc., Astronomy (*with Honors*) & B.Sc., Physics Cum Laude
Minors in Mathematics and Spanish
Advised by Prof. Nathan Smith
Attended Tecnológico de Costa Rica on exchange during Summer 2018

REFEREED PUBLICATIONS

2 first author, 5 Nth author, 1 first author in preparation, 2 second author in preparation, h-index = 7

1. **D. Dickinson**, D. Milisavljevic, J. Martin Laming, J. C. Raymond, R. Fesen, T. Temim, I. De Looze, S. Orlando, R. Arendt, B.-C. Koo, G. Katz 2026. High Resolution Mapping of the Unshocked Ejecta in Cassiopeia A. *In Prep.*
2. **D. Dickinson**, D. Milisavljevic, B. Garretson, L. Dessart, R. Margutti, R. Chornock, B. Subrayan, D. J. Hillier, E. Golib, D. Li, S. E. Logsdon, J. Rajagopal, S. Ridgway, N. Smith, C. Cynamon 2025. The Immediate, Exemplary, and Fleeting echelle spectroscopy of SN

2023ixf: Monitoring acceleration of slow progenitor circumstellar material, driven by shock interaction. ApJ. 10.3847/1538-4357/adc108

3. **D. Dickinson**, N. Smith, J. Andrews, P. Milne, C. D. Kilpatrick, D. Milisavljevic 2024. The Superluminous Type IIn Supernova ASASSN-15ua: Part of a Continuum in Extreme Precursor Mass Loss. MNRAS. 10.1093/mnras/stad3631
4. S. Orlando, H.-T. Janka, A. Wongwathanarat, **D. Dickinson**, D. Milisavljevic, M. Miceli, F. Bocchino, T. Temim, I. De Looze, D. Patnaude 2025. Filamentary Ejecta Network in Cassiopeia A Reveals Fingerprints of the Explosion Mechanism. A&A. 10.1051/0004-6361/202553833
5. I. De Looze, D. Milisavlejvic, T. Temim, **D. Dickinson**, R. Fesen, R. G. Arendt, J. Chastenet, S. Orlando, J. Vink, M. J. Barlow, F. Kirschlager, F. D. Priestley, J. C. Raymond, J. Rho, N. S. Sartorio, T. Scheffler, F. Schmidt, W. P. Blair, O. Fox, C. Fryer, H.-T. Janka, B.-C. Koo, J. M. Laming, M. Matsuura, D. Patnaude, M. Relaño, A. Rest, J. Schmidt, N. Smith, N. Sravan 2024. The Green Monster hiding in front of Cas A: *JWST* reveals a dense and dusty circumstellar structure pockmarked by ejecta interactions. ApJ 10.3847/2041-8213/ad855d
6. W. V. Jacobson-Galán, K. W. Davis, C. D. Kilpatrick, L. Dessart, R. Margutti, R. Chornock, R. J. Foleu, P. Arunachalam, K. Auchettl, C. R. Bom, R. Cartier, D. A. Coulter, G. Dimitriadis, **D. Dickinson**, M. R. Drout, A. T. Gagliano, C. Gall, B. Garretson, L. Izzo, D. O. Jones, N. LeBaron, H.-Y. Miao, D. Milisavljevic, Y.-C. Pan, A. Rest, C. Rojas-Bravo, A. Santos, H. Sears, B. Subrayan, K. Taggart, S. Tinyanont 2024. SN 2024ggi in NGC 3621: Rising Ionization in a Nearby, CSM-Interacting Type II Supernova. ApJ 10.3847/1538-4357/ad5c64
7. D. Milisavlejvic, T. Temim, I. De Looze, **D. Dickinson**, J. M. Laming, R. Fesen. J. C. Raymond, R. G. Arendt, J. Vink, B. Posselt, G. G. Pavlov, O. D. Fox, E. Pinarski, B. Subrayan, J. Schmidt, W. P. Blair, A. Rest, D. Patnaude, B.-C. Koo, J. Rho, S. Orlando, H.-T. Janka, M. Andrews, M. J. Barlow, A. Burrows, R. Chevalier, G. Clayton, C. Fransson, C. Fryer, H. L. Gomez, F. Kirschlager, J.-J. Lee, M. Matsuura, M. Niculescu-Duvaz, J. D. R. Pierel, P. P. Plucinsky, F. D. Priestley, A. P. Ravi, N. S. Sartorio, F. Schmidt, M. Shahbandeh, P. Slane, N. Smith, N. Sravan, K. Weil, R. Wesson, C. J. Wheeler 2024. JWST Survey of the Supernova Remnant Cassiopeia A. ApJ 10.3847/2041-8213/ad324b
8. B. M. Subrayan, D. Milisavljevic, R. Chornock, R. Margutti, K. D. Alexander, V. Ramakrishnan, P. C. Duffell, **D. Dickinson**, K.-S. Lee, D. Giannios, G. Lentner, M. Linvill, B. Garretson, M. J. Graham, D. Stern, D. Brethauer, T. Duong, W. Jacobson-Galán, N. LeBaron, D. Matthews, H. Sears, P. Venkatraman 2023. Scary Barbie: An Extremely Energetic, Long-duration Tidal Disruption Event Candidate without a Detected Host Galaxy at z = 0.995. ApJ 10.3847/2041-8213/accf1a
9. Z. Ding, **D. Dickinson**, D. Patnaude, D. Milisavljevic. The First Three-Dimensional Reconstruction of G292.0+1.8. *In Prep.*
10. B. Garretson, **D. Dickinson**, D. Milisavljevic, B. Subrayan, Q. Wang, R. Chornock, A. Rest, R. Margutti, G. Letner, M. Linvill, C. Cynamon. Tick, Tick... Boom! The Eruptive Countdown and Bimodal Rise of Type IIn Supernova SN 2023vbg. *In Prep.*

OBSERVING EXPERIENCE

Kuiper 61" Telescope	1 night
Bok 2.3-meter Telescope	1 night
MDM Observatory	3 nights

PRESENTATIONS

Invited Talks:

- Guest Lecture for Oak Forest High School, Oak Forest, IL, Dec. 2025
- Guest Lecture at Ivy Tech Community College PHYS 100 Class, Lafayette, IN, January 2026
- Big Boom Science Discussion, Tucson, AZ, October 2025
- Guest Lecture at Ivy Tech Community College PHYS 100 Class, Lafayette, IN, April 2025
- Purdue Astronomy Club Group Meeting, West Lafayette, IN, Oct. 2024
- Guest Lecture for Oak Forest High School, Oak Forest, IL, Oct. 2024
- Guest Lecture at Ivy Tech Community College PHYS 100 Class, Lafayette, IN, Aug. 2024
- V. A. Villar Group Meeting, Center for Astrophysics, Cambridge, MA, July 2024
- WIYN/NEID Board Meeting, April 2024
- CIERA/Northwestern Time-Domain Summer Journal Club, Aug. 2023

Contributed Talks:

- Rutgers Summer Transient Science Soiree, Piscataway, NJ, July 2025
- Rise_Time Conference, West Lafayette, IN Aug. 2024
- Spoken WERRD+ Transient Conference, Oct. 2022
- Big Boom Science Discussion, Tucson, AZ, Nov. 2020
- Purdue University Physics & Astronomy Final REU Mini-Conference, Aug. 2020
- Arizona/NASA Space Grant Consortium Annual Symposium, April 2020
- Big Boom Science Discussion, Tucson, AZ, Nov. 2019

Posters:

- 247th Meeting of the AAS, Phoenix, AZ, Jan. 2026
- Purdue Physics Poster Session, Wes Lafayette, IN, Nov. 2025
- 245th Meeting of the AAS, National Harbor, MD, Jan. 2025
- Purdue Physics Poster Session, West Lafayette, IN, Nov. 2024
- Supernova Remnants III, Chania, Greece, June 2024
- WISP Travel Grant Symposium, West Lafayette, IN, April 2024
- 243rd Meeting of the AAS, New Orleans, LA, Jan. 2024
- 241st Meeting of the AAS, Seattle, WA, Jan. 2023
- 237th Meeting of the AAS, Jan. 2021
- 235th Meeting of the AAS, Honolulu, HI, Jan. 2020
- University of Arizona First Year Project Symposium, Tucson, AZ, April 2018

HONORS & ACHIEVEMENTS

2026	Philanthropic Educational Organization (PEO) Scholar Award
2024	Graduate Student Chambliss Award
2023	The Gabriele F. Giuliani Award
2022	Rolf Scharenberg Graduate Summer Research Fellowship
2021	Outstanding Senior for the Department of Astronomy and College of Science Finalist
2020-2021	Mortar Board National College Senior Honor Society
2020-2021	Galileo Circle Scholar

2020-2021	Slipher Scholarship
2017-2021	Academic Distinction
2019-2020	University of Arizona/NASA Space Grant Internship
2017-2021	Arizona Excellence Scholarship
2017-2021	Academic Recognition Award

TEACHING, MENTORSHIP, AND VOLUNTEER EXPERIENCE

Physics Graduate Student Association Board Member *2023 - 2026*

- President 2025 - 2026
- Vice President 2024 - 2025
- Secretary 2023 - 2024
- Secretary 2022-2023

SMAP AstroCamp in Medellin, Colombia *March 2023, March 2026*

- Co-hosted with Colegio San José de las Vegas, Universidad EIA, and Planetarium La Enseñanza
- Three-day astrophysics camp in 2026 covered the Doppler effect, exoplanets, crater physics, particle physics.
- Four-day astrophysics camp in 2023 covered stellar evolution, massive stars, supernovae, supernova remnants, and cosmology

Teaching Assistant for PHYS 323/324 (Data Mine) *Fall 2022, Spring 2026*

- Mentored and oversaw the research of Manuel Gil, Morgan O'Keefe, Charlee Shubert, and Erin Smith during Fall 2022
- Mentored and oversaw the research of Jason Emsley, Chawin Mingsuwan, Jing Gan, Jay Gannam, Honghai Gong, and Francisco Ruiz during Spring 2026

Graduate Teaching Assistant *Fall 2021 - Spring 2023, Spring 2026*

- PHYS 323/324: Research with Big Data I/II
- PHYS 450: Optics Laboratory I
- ASTR 263/264: Descriptive Astronomy: The Solar System/Stars & Galaxies
- PHYS 234: Physics for Life Sciences II

Purdue Undergraduate Research Expo Judge *Spring 2023 - Fall 2024*

Undergraduate Teaching Assistant *Spring 2021*

- Class TA for Physics 331: Electricity and Magnetism I
- Responsible for grading and Formal Reports
- Taught LaTeX to students

Preceptorship *Fall 2018*

- Class learning assistant for Astronomy 170B: The Physical Universe

MEDIA AND PUBLIC OUTREACH

2021-Present	Saturday Morning Astrophysics Lecturer and Teaching Assistant
2026	SMAP AstroCamp 2026 Lecturer and Teaching Assistant*
2025	Public Talk, Astronomy on Tap: The Hidden Art in Astrophysics
2025	Emcee for the Purdue Vera Rubin Observatory First Look Watch Party
2024	Appearance in Purdue Impact Research Vignette on the Big Ten Network
2023	Appearance in a JWST Edition of CNN's "The Whole Story" with Anderson Cooper*
2023	Purdue Physics & Astronomy Student Spotlight*
2023	SMAP AstroCamp 2023 Lecturer and Teaching Assistant*
2022	Public Talk, Astronomy on Tap: Vampires and Werewolves in Space!
2022	Co-emcee for Purdue's JWST First Light Discussion*
2022	Purdue Physics & Astronomy Instagram Takeover
2021	Graduate Student Highlight* with Purdue Physics & Astronomy
2021	Undergraduate Student Highlight* with UArizona Research

**Links to these can be found at https://linktr.ee/danielle_sci_ed*

EMPLOYMENT HISTORY

Lafayette Jewish Community Religious School Teacher
Supervisor: Dr. Eliza Gellis. *Fall 2025 - Present*
Purdue University, West Lafayette, IN.

Smithsonian Astrophysical Observatory Visiting Fellow
Supervisors: Dr. John Raymond & Dr. Dan Patnaude. *Summer 2024*
Center for Astrophysics, Cambridge, MA

- Advised by Dr. John Raymond in contributing to my PhD Thesis.
- Reduced and analyzed long-slit spectroscopy of G292 in contributing to the PhD Thesis of Ziwei Ding.

Graduate Research Assistant
Supervisor: Prof. Dan Milisavljevic. *Summer 2022, May 2023 - Present*
Purdue University, West Lafayette, IN.

Graduate Teaching Assistant
Supervisors: Prof. Adrzej Lewicki, Lucia Nelson. *August 2021 - May 2023, Spring 2026*
Purduer University, West Lafayette, IN.

Astronomy Camp Counselor
Supervisor: Prof. Don McCarthy. *Summer 2021*

- Led groups of middle and high school students in Zoom lectures, interactive learning activities, and virtual observing runs.

Science Interpreter & Planetarium Operator
Supervisor: Noel Hensley. *April 2019 - May 2021*
Flandrau Science Center and Planetarium, Tucson, AZ.

- Led groups of K-12 students and engage them in the exhibits and hands-on learning.
- Operated the planetarium theatre and give public lectures after shows on astrophysics and other scientific topics.

Assistant Baker and Chef
Supervisor: Alaine Shrewsbury. *April 2018 - December 2018*
Green Heart Bakery, Tucson, AZ.

- Assisted Alaine Shrewsbury in cooking and baking food items for sale at the University of Arizona and Rillito farmers markets.

NON-REFEREED PUBLICATIONS

Weil, K. E., et al. (incl. **Dickinson. D**). REFITT classifications of optical transients using SOAR. Transient Name Server AstroNote 222, 1-222.

Weil, K. E., et al. (incl. **Dickinson. D**). REFITT classification of SN 2021nxq (ZTF21abcpsjy). Transient Name Server AstroNote 182, 1-182.

Weil, K. E., et al. (incl. **Dickinson. D**). REFITT classifications of optical transients using SOAR. Transient Name Server AstroNote 30, 1-30.

Sravan, N., et al. (incl. **Dickinson. D**). Autonomous real-time science-driven follow-up in the era of LSST. AAS 237, 211.05.

Weil, K. E., et al. (incl. **Dickinson. D**). REFITT classifications of optical transients using SOAR. Transient Name Server AstroNote 266, 1.

Weil, K. E., et al. (incl. **Dickinson. D**). REFITT Discovery and Classification of SN2020abog (ZTF20acpgjac) using SOAR. Transient Name Server AstroNote 243, 1.

Weil, K. E., et al. (incl. **Dickinson. D**). REFITT classifications of optical transients using SOAR. Transient Name Server AstroNote 242, 1.

Weil, K. E., et al. (incl. **Dickinson. D**). REFITT classifications of optical transients using SOAR. Transient Name Server AstroNote 232, 1.

Weil, K. E., et al. (incl. **Dickinson. D**). REFITT Discovery and Classification of SN 2020zct (ZTF20acezhcf) using SOAR. Transient Name Server AstroNote 227, 1.

Weil, K. E., et al. (incl. **Dickinson. D**). REFITT classifications of optical transients using SOAR. Transient Name Server AstroNote 225, 1.